Industrial Marketing Management 71 (2018) 27–40

Contents lists available at ScienceDirect

Industrial Marketing Management

journal homepage: www.elsevier.com/locate/indmarman

Explicating the relationship of entrepreneurial orientation and firm performance: Underlying mechanisms in the context of an emerging market

Lin Cui[a], Di Fan[b,*], Feng Guo[a,c], Yi Fan[a]

[a] Research School of Management, College of Business and Economics, Australian National University, Canberra, Australia
[b] School of Management, Curtin Business School, Curtin University, 408.4031, Kent St. Bentley, Perth, WA 6102, Australia
[c] School of Management, Harbin Institute of Technology, 92 West Dazhi Street, Nan Gang District, Harbin 150001, China

ARTICLE INFO

Keywords:
Entrepreneurial orientation
Absorptive capacity
Boundary-spanning
Institutional voids
Market-supporting institutions
Emerging market

ABSTRACT

This research examines the mechanisms through which entrepreneurial orientation (EO) affects firm performance in an emerging market context. We argue that EO drives firms to develop the dynamic capabilities of absorptive capacity (ACAP) and boundary-spanning (BS), and that these capabilities enhance firm performance. Moreover, the effectiveness of these knowledge- and network-based dynamic capabilities in mediating the EO-performance relationship is contingent on the institutional conditions firms are exposed to. We test our hypotheses using multi-sourced data on Chinese high-tech firms in business-to-business (B2B) markets. Our results show that ACAP and BS mediate the positive effect of EO on firm performance, and that as the development of market-supporting institutions improves, the mediating effect of ACAP becomes stronger while that of BS weakens.

1. Introduction

Entrepreneurial orientation (EO) reflects the strategic posture of a firm to continuously engage in risk-taking, innovative, and proactive behaviours which will ultimately influence firm performance (Covin & Slevin, 1989; Hoskisson, Covin, Volberda, & Johnson, 2011). While the antecedents, outcomes, and performance contingencies of EO have been extensively researched, theoretical development and empirical investigation regarding the generative mechanisms underlying the EO-performance relationship are currently lacking (Wales, Gupta, & Mousa, 2013). It remains unclear what capabilities firms generate from their EO which in turn enable them to attain positive market performance. Understanding these generative mechanisms requires the examination of the mediators that play a pivotal role in delivering the performance effect of EO (Lumpkin & Dess, 1996; Wales, Gupta, & Mousa, 2013). These mechanisms and their contingencies carry significant business implications for prioritizing resources to the capability developments that deliver performance outcomes for entrepreneurial firms. In this study, we address this knowledge gap by asking: *What are the capability development mechanisms that allow firms to convert their EO into performance outcomes? What environmental contingencies promote/hinder such mechanisms for entrepreneurial firms?*

Recent studies have attempted to identify the missing link between EO and firm performance. For instance, Kollmann and Stöckmann (2014) examined exploratory innovation and exploitative innovation as mediating variables within the EO-performance relationship. Similarly, other studies have also investigated learning related constructs, such as information acquisition and utilization, to reveal the mechanisms mediating the EO-performance relationship (Keh, Nguyen, & Ng, 2007; Wang, 2008). These studies offer insight into the *knowledge-based capability* through which entrepreneurial firms create value. They suggest that firms with risk-taking, innovative, and proactive behavioural tendencies are likely to seek novel knowledge from their dynamic environments to improve their processes and product offerings, which in turn enhances and sustains their competitiveness.

Another stream of EO research focuses on the institutional context within which entrepreneurial firms operate (Covin & Miller, 2014; Dess, Pinkham, & Yang, 2011). This stream of research highlights that when information and market transactions are not sufficiently facilitated by market-supporting institutions, firms with non-market channels to access critical information and resources will have a competitive advantage over those that do not (Peng & Luo, 2000). Such business environments are commonly found in, but not limited to, emerging markets, where the underdevelopment of market-supporting institutions presents "institutional voids" (Khanna & Palepu, 1997), leaving profit opportunities for those entrepreneurial firms who can effectively integrate and leverage network resources, such as social capital (Covin & Miller, 2014), to bridge these voids. This *network-based capability* of

* Corresponding author.
E-mail addresses: lin.cui@anu.edu.au (L. Cui), david.fan@curtin.edu.au (D. Fan), guofeng229@hit.edu.cn (F. Guo), u4713133@anu.edu.au (Y. Fan).

https://doi.org/10.1016/j.indmarman.2017.11.003
Received 2 August 2015; Received in revised form 7 September 2017; Accepted 6 November 2017
Available online 21 November 2017
0019-8501/

value creation by entrepreneurial firms has been relatively under-explored in the EO literature.

This study aims to provide an integrated understanding of the capability development mechanisms underlying the performance effect of EO. Through integrating dynamic capabilities and institutional perspectives, we identify the development of knowledge-based and network-based dynamics capabilities driven by EO, and reveal the institutional contingencies that promote or hinder the development of such capabilities. We argue that a high EO drives firms to upgrade their processes and/or products by engaging in external searches for novel knowledge using the dynamic capability of knowledge acquisition, assimilation, transformation, and exploitation, formally known as ACAP (Zahra & George, 2002). Meanwhile, in an emerging market context, an entrepreneurial firm can also seek to differentiate itself by integrating and leveraging network resources to bridge institutional voids and to seize innovation and business opportunities that are unavailable to its competitors (Xin & Pearce, 1996). This method of achieving entrepreneurial success through network connections highlights the dynamic capability of boundary-spanning (BS) (Aldrich & Herker, 1977; Dollinger, 1984). Furthermore, the relative effectiveness of ACAP and BS as dynamic capabilities that transmit EO into competitive advantages is contingent on institutional development in a firm's operating environment, such that entrepreneurial firms will prioritize the development of ACAP over BS when market institutions are better developed.

Empirically, this study uses multi-sourced data from 411 Chinese firms to test the theorized moderated mediation relationships in an emerging market business-to-business (B2B) context. We focus on small-to-medium sized enterprises (SMEs) operating in four high-tech industrial clusters embedded in sub-national locations with varying degrees of marketization. We combine multi-respondent survey data with archival data in our analyses using the structural equation modelling (SEM) technique. We find strong support for the hypothesized mediating effects of ACAP and BS that link EO to positive performance outcomes. We also find that, as hypothesized, a high level of marketization in a firm's institutional environment strengthens the mediating effect of ACAP and weakens the mediating effect of BS.

Overall, this study makes three contributions to the EO literature. First, we shed new light on the underlying generative mechanisms associated with EO, which has been a significant knowledge gap in the extant literature (Wales et al., 2013). We link EO as a behavioural construct (Covin & Lumpkin, 2011) to dynamic capability development as a mechanism of performance achievement, where we identify two types of capabilities, which are knowledge-based and network-based respectively. Second, we deepen the understanding of the institutional contingencies on the EO-performance relationship. We find that the mediated performance effect of EO through ACAP and BS is moderated by the external institutional environment, which suggests that the effectiveness of capability development towards entrepreneurial success is institutionally contingent. This finding has important resource allocation implications for entrepreneurial firms. Third, we extend EO research into a novel empirical context – B2B firms in an emerging market. Our empirical findings further strengthen EO as a useful construct for explaining business success in a variety of institutional and industrial contexts.

2. Theory and hypotheses development

Entrepreneurial firms gain competitive advantages through their capabilities in leveraging exclusive information and resources that are not available to competitors (Shane & Venkataraman, 2000). Research suggests that firms may achieve entrepreneurial advantages in two ways. First, firms can create new and unique resource bundles that are not available to rival firms, and then benefit from the exclusive understanding about the value of these newly created bundles, before they can be imitated by competitors. Creating new resource bundles requires firms to engage in acquisition, assimilation, transformation, and exploitation of external knowledge through a set of organizational routines and processes reflective of a dynamic capability known as absorptive capacity or ACAP (Zahra & George, 2002). Second, market interventions by government or powerful economic actors (e.g. market monopolies and business groups) can create competitive barriers that hinder the free flow of information and efficient transactions in both factor and product markets. Accordingly, firms can obtain information and resource advantages by penetrating those barriers (Peng & Heath, 1996), which requires firms to establish and utilize network-bridging social capital through managerial capabilities characterized by boundary-spanning or BS (Cao, Simsek, & Jansen, 2015; Park & Luo, 2001).

In an emerging market context, both the abovementioned dynamic capabilities (i.e., ACAP and BS) are viable channels for entrepreneurial firms to achieve superior performance. Market liberalization exposes emerging market firms to global competition, which enhances their competitive awareness and motivates them to engage in vicarious learning and the acquisition of strategic assets. ACAP enables emerging market firms to not only simply "copycat" foreign firms, but to also effectively identify complementary assets and integrate them with their existing competencies to create unique and valuable new resource bundles (Luo, Sun, & Wang, 2011). When institutional development unfolds gradually, market institutions remain largely underdeveloped in major emerging economies, which creates institutional voids (Khanna & Palepu, 1997, 2010; Tracey & Phillips, 2011). Firms that are able to span network boundaries to overcome such institutional voids are able to secure substantial competitive advantages (Xin & Pearce, 1996). Recognizing the development of these two types of dynamic capabilities as alternative processes through which entrepreneurial firms create a competitive edge, we propose two parallel mediating mechanisms that link EO with performance outcomes via ACAP and BS respectively. Furthermore, we argue that the effectiveness of the knowledge-based (i.e., ACAP) and network-based (i.e., BS) dynamic capabilities is likely to vary depending on the institutional environment in which the entrepreneurial firm operates. As market-supporting institutions develop, the value of relational assets decreases while that of market-oriented competitive resources, such as business know-how and technology, increases (Peng, 2003). Therefore, the mediating effects of ACAP and BS are differentially moderated by market institutional development, as entrepreneurial firms perceive a different value in investing in these alternative dynamic capabilities.

Next, we elaborate on the arguments for the two parallel mediating mechanisms and their institutional contingency which led to our research hypotheses. In addition to theoretically derive these hypotheses, we conducted in-depth qualitative case studies to contextualize our arguments. Four case firms were selected, based on their representativeness of the industry, (see Appendix A for details of the case studies) for in-depth semi-structured interviews with senior executives. The interviews took place between June 2016 and April 2017. We followed the practice of Pahnke, Katila, and Eisenhardt (2015) to integrate qualitative information in our hypothesis development.

2.1. The mediating role of ACAP

ACAP was originally defined as the "ability to recognize the value of new information, assimilate it, and apply it to commercial ends" (Cohen & Levinthal, 1990: 128). Zahra and George (2002: 186) extend this definition to add that ACAP is "a set of organizational routines and processes by which firms acquire, assimilate, transform, and exploit knowledge to produce a dynamic organizational capability". We argue that developing ACAP is a viable channel for entrepreneurial firms to obtain competitive advantages as reflected in superior market performance for two main reasons.

First, by definition, ACAP as a dynamic capability allows firms to maintain competitively distinctive in fast-changing technological and market environments, which will result in superior firm performance. However, not all firms are equally positioned to develop ACAP. The path-dependent nature of ACAP suggests that this dynamic capability is a function of a firm's consistent pursuit of novel knowledge (Zahra & George, 2002). EO, as a strategic posture, reflects the inclination of a firm to engage in sustained behaviours that are risk-taking, innovative, and proactive in nature (Covin & Lumpkin, 2011; Miller, 1983). When engaging in these behavioural patterns, firms with high EO are better positioned to develop ACAP, because organizational searches for novel external knowledge often involves high risk and uncertain returns in exchange for potential innovative and proactive solutions to technological and operational challenges. Risk-taking tendencies motivate firms to tap into sources of distant and unfamiliar external knowledge, and this has a high potential to lead to innovative changes (Lisboa, Skarmeas, & Lages, 2011). Similarly, proactive firms tend to focus on developing new competencies, as opposed to maintaining the status quo, which requires those firms to explore and leverage novel knowledge externally. Persistent searches for, and utilization of, external knowledge contributes to the development of ACAP (Zahra & George, 2002).

Second, ACAP as a knowledge-based dynamic capability enables firms to identify and capture entrepreneurial opportunities. A high level of ACAP allows firms to access broader information from external sources and to effectively assimilate the information to identify its value. Such an information advantage not only better positions firms to anticipate future technological changes and market needs, but also provides firms with the capacity to act on those future needs through proactive innovation (Miller, 1983; Miller & Friesen, 1978). If firms are able to transform the externally acquired knowledge in combination with existing internal routines and practices (cf. Chen, Li, & Evans, 2012), then they are likely to synergize on experiential and vicarious learning and, as a result, create new and unique resource bundles that are unknown and/or unavailable to competitors. Overall, ACAP contributes to organizational learning and the development of sustainable competitive advantage, which in turn leads to the performance improvement of firms (Tsai, 2001; Zhang, Wu, & Henke, 2015). The funding CEO of Shifang, an automation engineering product manufacturer, described the process to us in this way: "*Absorptive capacity has benefited my career growth and our business operation. Since I established this automation engineering work firm, I have kept emphasising the importance of knowledge-seeking and our own R&D ability to all engineers, simply because I deeply believe how important they were for our firm performance in the past.*" For example, a key B2B product of Shifang is an industrial controlling system, and an important component of this is called the Digital Frequency Converter, of which the key technology was held by German giant ABB and Japanese company Sanyo Electric. The CEO added: "*As you can see, this is a process that we learnt and introduced foreign technology to local markets. Apart from this, we also partner with other high-tech automation firms in Beijing, Tianjin, and Nanjing …we strongly encourage our engineers to master new technology then develop our own knowledge pool. We call this process learning-by-doing, and learning-by-using.*"

In summary, a high EO better positions firms to develop ACAP (Wiklund & Shepherd, 2003), and, in turn, enhanced ACAP will lead to better firm performance (Chen, Lin, & Chang, 2009; Tsai, 2001). As such, ACAP is an important mediator linking EO with firm performance:

H1. The absorptive capacity of an emerging market firm positively mediates the effect of entrepreneurial orientation on performance.

2.2. The mediating role of BS

BS refers to the establishment and utilization of social ties with the external entities on which a focal firm depends (Geletkanycz & Hamcrick, 1997). From an institutional perspective, firms in underdeveloped institutional environments rely more on relationships (e.g. social ties and networks) than on formal market institutions to negate the impact of environmental uncertainties and turbulence, as well as to identify and seize entrepreneurial opportunities (Peng, 2003). Important business processes, such as obtaining market information, interpreting regulations, and enforcing contracts, may rely heavily on network connections in the emerging market context (Peng & Luo, 2000). Therefore, building social capital to leverage institution-substituting networks plays an important role in accessing critical external buffers and overcoming institutional voids (Xin & Pearce, 1996). In particular, when underdeveloped institutions increase the costs for information searching, financing, and business transactions, businesses rely on the managerial capabilities of spanning network boundaries with government, business partners, and research institutes to access critical information, as well as technological and financial resources (Park & Luo, 2001; Peng & Luo, 2000). Accordingly, BS represents a network-based dynamic capability that grants firms, especially those operating in emerging markets, advantageous access to information and resources that are too costly for or unavailable to competitors. As an alternative to the development of knowledge-based dynamic capability (i.e., ACAP), BS is also a viable channel through which entrepreneurial firms can attain superior performance, for the following two main reasons.

First, a high level of EO will motivate firms to explore and leverage network resources to support their risk-taking, innovative, and proactive activities. Risk-taking incurs resource commitments to business opportunities despite a reasonable chance of a costly failure in market terms (Lumpkin & Dess, 1996; Miller & Friesen, 1978). As such, firms need to draw on resource buffers from non-market channels. For instance, studies show that Chinese firms with political connections are more able to access resource buffers by the state that will assist them to manage risks (Buckley et al., 2007; Zheng, Singh, & Mitchell, 2015). A high EO is also reflected in a firm's pursuit of business innovation to develop novel and creative resource bundles. Thus, firms with high aspirations for innovation are motivated to establish and actively utilize multiple networks, as opposed to solely relying on factor markets, so that they can potentially benefit from network arbitrage to obtain innovative resources, and they can also minimize a network "locked-in" effect that can drive firms towards a central tendency (Geletkanycz & Hambrick, 1997). Moreover, firms with a high EO are, by definition, proactive in anticipating and acting on future market trends (Lumpkin & Dess, 1996; Miller & Friesen, 1978). The analysis of future market trends requires high quality and diverse information about technological, regulatory, social, and other environmental changes, which are often disseminated through different network channels. As a result, information asymmetry is created based on network memberships. Firms with high proactiveness are therefore motivated to gain memberships to important networks which have access to valuable information (cf. Zheng, Singh, & Mitchell, 2015). Occupying asymmetric information allows firms to identify business opportunities and take action ahead of potential rivals. The need to engage in BS to support entrepreneurial activities is reflected on by the funding CEO of Zhonghe Tech., an online platform technology provider: "*When we started this high-tech firm, we had really tough period in terms of finding venture capital, obtaining governmental approval, developing network technology, and studying our potential customers etc. All these require social ties*". Regarding one of this company's key product development projects (an online group coupon purchasing platform) the CEO elaborated on their efforts in sourcing information and funding from external networks: "*We did not only need our technology team and external experts who can devote themselves to implement core system, but also require strong relationships with marketing agencies who can analyse our industrial partners' needs. As a boss, I was also busy for finding money to finance this project by utilising all possible network channels and personal connections, which I think is vital for the survival of internet company like us*".

Second, the development of network-based dynamic capabilities (i.e., BS) motivated by a high level of EO can lead to an advantageous market position for achieving superior performance. When facing institutional voids, network-bridging social capital allows firms to obtain resources and market legitimacy in a more cost-effective way than through default market channels (Peng & Luo, 2000; Xin & Pearce, 1996). For example, social capital with administrative and regulatory bodies are critical to business operations in emerging markets. Firms build government ties for various purposes, such as to compensate for incomplete formal institutions, to overcome institutional barriers, to adapt to regulatory changes, and to obtain policy support, project approval, and state-owned scarce resources (Luo, 2003; Park & Luo, 2001; Xin & Pearce, 1996). Government ties can reduce the institutional costs for firms as the firms pursue entrepreneurial activities, as firms can utilize such informal institutions (i.e., relational networks) to gain legitimacy. Through their connections with the government, firms can also gain valuable insights into policy changes that may later lead to economic disequilibrium with regard to certain resources and thus provide valuable entrepreneurial opportunities. Furthermore, access to government financing channels, such as state-owned banks, can provide firms with capital resources to successfully pursue entrepreneurial activities. In addition to official networks, BS firms can also leverage other types of networks to obtain a competitive market position. For instance, firms cultivate business networks for competitive purposes, such as to obtain customers (Peng & Luo, 2000), utilize market opportunities and counter market threats (Carroll & Teo, 1996; Geletkanycz & Hambrick, 1997), secure credit, open dialogues, and build trust (Hoskisson, Eden, Lau, & Wright, 2000). In the case of Zhonghe Tech., the critical role of BS as a dynamic capability for business success is recognized by the CEO: "*Internet technology has been developed so rapidly; without strong networking abilities to source all kinds of resources, we won't have a chance to catch up technology updates and develop business ideas*".

In summary, a high EO motivates firms to engage in BS to obtain network-based competitive advantages, which will then be reflected in their superior performance. The above arguments suggest that BS functions as an alternative channel through which EO impacts firm performance. Accordingly, we propose:

H2. Boundary-spanning of an emerging market firm positively mediates the effect of entrepreneurial orientation on firm performance.

2.3. The moderating effect of market-supporting institutions

As discussed above, EO-induced ACAP and BS represent two main channels through which firms generate superior performance from their entrepreneurial behaviours. While both mediating channels are viable alternatives for entrepreneurial firms, their relative effectiveness is likely to be contingent on the institutional environment in which a firm operates, because different types of resources (e.g., technology vs. social capital) are likely to vary in value when the nature of business transactions change (Peng, 2003), especially as emerging market countries undergo institutional changes towards the marketization of their economies (Banalieva, Eddleston, & Zellweger, 2015).

Emerging economies are frequently experiencing institutional transitions (Peng & Heath, 1996), which represent deliberate shifts in core sets of formal institutions, typically the development of market-supporting institutions. Market-supporting institutions facilitate economic transactions by providing unbiased price signals and contract enforcement (North, 1990; Williamson, 1992). They consequently enable market transactions for the efficient allocation of economic resources. As the principal "rules of the game" for inter-firm relationships, they determine which types of resources are likely to deliver competitive distinctiveness. Specifically, Peng (2003) argues that during market-oriented institutional transitions, relationship-based transactions decline while rule-based transactions emerge and gradually gain dominance. Due to this shift in transaction types, the sources of competitive advantage change from relationship and networks to competitive resources and capabilities (Peng, 2003).

Accordingly, entrepreneurial firms exposed to different levels of market-supporting institutions are likely to find the potential effectiveness of ACAP versus BS channels of competitiveness building varies. As such, they will allocate organizational resources to these channels differently. ACAP as a knowledge-based dynamic capability supports the development of firm-specific competitive advantages by accessing and utilising novel knowledge. Knowledge access is most efficient when information exchange in the factor and product markets are facilitated by market intermediaries and are not obstructed by regulatory barriers. The exploitation of novel knowledge for innovation also requires a value-expropriation regime where business transactions and competition are rule-based (Zahra & George, 2002). However, when market-supporting institutions are underdeveloped, lack of market intermediaries makes information exchange costly and unreliable, and this compromises the input process of ACAP. Also, the output process of ACAP is compromised when intellectual property is not effectively protected so that firms can profit from their innovation efforts. As such, the efficacy of ACAP is conditional on market institutional development. Our interviews with the general manager and a co-founder of BLB Biology, a corn-based health and food products manufacturer, revealed that ACAP had been an important component in achieving business success: "*Deep and broad learning to enhance technology competitiveness has been embedded in our corporate culture and tradition*". The manager was, however, highly cognizant of the boundary condition for this capability: "*It wouldn't be possible to establish our business without market economy as a national policy*". He further elaborated: "*Once having better market-supporting policies offered by local governments, we are able to more efficiently access knowledge from upstream and downstream business partners, and to have the value of our innovative products appropriately recognized by the market, rather than having our product price regulated by bureaucrats*". Overall, the effectiveness of ACAP as a channel of transmitting EO to competitive advantage is positively associated with market-supporting institutions. In other words, firms with high levels of EO are more likely to achieve high performance through the development of ACAP when they operate in environments with more developed market-supporting institutions.

H3. a: The development of market-supporting institutions in a firm's operating environment positively moderates the indirect effect of EO on firm performance mediated by ACAP.

b: The development of market-supporting institutions in a firm's operating environment negatively moderates the indirect effect of EO on performance mediated by BS.

In contrast to ACAP, BS as a network-based dynamic capability creates value in the form of social capital that allow firms to bridge network barriers and access information and resources not available on the open market. Social capital has contingent value. It is more valuable when networks perform a greater economic coordination role than does the market, especially when market functions, such as price mechanism and contract enforcement, are undermined by institutional voids (Khanna & Palepu, 1997, 2010; Tracey & Phillips, 2011). For instance, the literature on business groups suggest that, under institutional voids, inter-firm networks enable firms to share, transfer, and exploit various resources more efficiently through internal markets, which in turn deliver competitive advantages (Chang, Chung, & Mahmood, 2006). Moreover, weak market-supporting institutions are often accompanied by strong state coordination of the economy, where the government actively intervenes in the economy by shaping the supply of economic inputs and regulating business activities. Given the significance of state influence, firm performance is likely to be dependent on the network capabilities required to manage relationships with the government (Holburn & Vanden Bergh, 2008; Mahon & Murray, 1981). However, as market-supporting institutions develop, the state defers economic coordination to the market (Peng, 2003), and the emergence of market infrastructure for innovation dilutes the advantages of networks (Chang, Chung, & Mahmood, 2006). The contingent value of social capital for supporting entrepreneurial activities is demonstrated in the case of Jinru, a renewable energy technology provider with its headquarters in Beijing. Jinru operates its power generation plant in Inner Mongolia, an inland province of China where the development of market institutions lags significantly behind that of the national capital Beijing. The founder of Jinru said: "*Unlike doing business in Beijing, we needed to build strong relations with local governments here [Inner Mongolia] because officials can help exempting or quickening all necessary approval procedures; and we also needed to cultivate good relationship with local farmers and technicians as their way of doing business is very much personal and trust-based, unlike the formal rules and regulations we are used to in major cities* [where market-institutions are more developed]". This entrepreneur also had business ventures in information technology and property management located in the more marketized provinces of China. When asked if relational ties were also critical for venture success in these locations, he replied immediately: "*No, relationships are less useful in more competitive markets. Relationship per se will not help you much if there are clear rules to follow and everyone plays by the market rules*". Accordingly, the effectiveness of BS as a channel of transmitting EO to competitive advantage is negatively associated with market-supporting institutions. In other words, firms with high levels of EO are less likely to achieve high performance through the development of BS when they operate under more developed market-supporting institutions.

3. Methods

3.1. Sample and data collection

We tested our hypotheses in the context of the Chinese economy, which presents a great variety of entrepreneurial activities from local firms aiming to advance their positions by leapfrogging their competitors within challenging institutional and competitive environments (Luo, Sun, & Wang, 2011; Tang, Tang, Marino, Zhang, & Li, 2008). China also has significant within-country variations of institutional development, due to the continuous institutional transitions under heterogeneous economic and socio-cultural conditions across sub-national regions (Fan, Wang, & Zhu, 2011; Wang, Wong, & Xia, 2008). Examining marketization at sub-national levels instead of through cross-country comparison reduces the potential for any confounding effects based on between-country differences, such as cultural and political differences.

We focused on SMEs in high-tech B2B markets where entrepreneurial activities are arguably more vibrant and critical to the survival and success of firms than they are to firms in consumer markets (Ulaga & Sharma, 2001; Yang & Gabrielsson, 2017). Our sampling frame consisted of four long-established high-tech clusters, namely, Beijing Zhongguancun Science & Tech Park, Jinan Software Cluster, Baotou SME Venture Park, and Shenzhen High-tech Cluster. Firms operating in these high-tech clusters focus on the supply of intermediate products and services to supply chain partners (Zhang & Li, 2010). A total of 20,209 firms were identified using the company lists obtained from the administrative committees of the four clusters. A randomized sample of 2000 firms was used for this study.

We combined survey and archival data to construct a multi-sourced dataset. The survey was conducted in the first half of 2011. To prevent common method bias, the questionnaire was designed in two parts, separating the key constructs involved in the hypothesized relationships. Specifically, general firm strategic posture (e.g. EO) and performance were evaluated by the chief executive officer, general manager, or chairperson of the firm who were invited to respond to part one of the survey, whereas information related to firm capabilities (e.g. ACAP and BS) was included in part two for which the chief technology officer or the chief information officer was the desired respondent. The initial questionnaire was designed in English and translated into Chinese by two bilingual academics with expertise in entrepreneurial activities. The questionnaire was reverse-translated for accuracy and pre-tested in

Table 1
Description of sample firms and respondents.

Firm size (employees)		Industry		Firm ownership type	
Up to 10	30	Information technology	107	State-owned or controlled	43
11 to 50	160	Bio technology	47	Private-owned	318
51 to 100	83	New energy	49	Collective-owned	48
101 to 150	44	New material	48	Mixed ownership type	18
151 to 200	32	Environment	45		
> 200	62	Electronic	58	**Location**	
		Creative industry	28	Cluster 1 (Beijing)	152
Firm age		Other industries	29	Cluster 2 (Jinan)	62
Up to 4 years	25			Cluster 3 (Baotou)	78
5 to 12 years	62	**Founder type**		Cluster 4 (Shenzhen)	119
13 to 20 years	89	Overseas returnee	70		
> 20 years	235	Local founder	341	**Respondent education**	
				High school or below	17
		Respondent position		Bachelors	307
		CEO/general manager	411	Masters (including MBA)	312
		CIO/CTO	411	Doctorate	186

Note: N = 411, two respondents each firm.

an EMBA class of Chinese managers to ensure the meanings of the questions were easily understandable. A total of 411 firms returned the fully completed questionnaires, constituting an effective response rate of 20.6%, which is comparable to the response rates of executive surveys in prior studies. Sample descriptions are reported in Table 1.

We performed non-response-bias tests to detect any self-selection issues. We collected information on firm size, age, state ownership, and industry concentration ratio using secondary data obtained from the administrative committees of the clusters. *t*-tests between respondent and non-respondent firms did not return any significant results. We also followed Heckman's two-stage procedure and estimated a probit model of selectivity on identifying variables including firm size, age and state ownership. None of the variables were significant and the probit model result was also non-significant. The Inverse Mills' ratio (lambda) calculated from this procedure was also non-significant when entered in a regression model of composite performance. Based on the results of these procedures, we concluded that there was no evidence of self-selection issues.

In addition to the survey, we collected archival data on the marketization index of the operating environment of each sample firm. Marketization indexes are sourced from the National Economic Research Institute's (NERI) marketization index of Chinese provinces (Fan, Wang, & Zhu, 2011), which is widely used for strategy research in the Chinese context (Banalieva, Eddleston, & Zellweger, 2015; Shi, Sun, & Peng, 2012; Wang, Wong, & Xia, 2008).

3.2. Variables and measurements

The key variables in our model are latent constructs. Table 2 presents the measurement scales, consulted references, and the results of a confirmatory factor analysis (CFA) of the measurement scales using SEM. We used Cronbach's alpha and Hancock and Mueller's coefficient H to assess construct reliability.

The dependent variable was firm *performance*. This variable has been commonly measured using a four-item scale, with the items being profitability, market share, sales, and overall financial performance. The four-item scale is based on self-rated firm performance relative to principle competitors over the previous three years (Keh, Nguyen, & Ng, 2007; McDougall, Covin, Robinson, & Herron, 1994; Tang, Tang, Marino, Zhang, & Li, 2008). This four-item scale returned high construct reliability ($\alpha = 0.91$; $H = 0.88$).

Table 2
Measurement scales and reliabilities.

Constructs and indicators	Factor loading	t	R^2	Reliability	
				H	α
Performance (Keh et al., 2007; McDougall, Covin, Robinson, & Herron, 1994; Tang et al., 2008)				0.88	0.911
Profit growth rate before tax	0.78	33.70	0.61		
Market share	0.77	32.20	0.59		
Sales growth rate	0.83	42.16	0.69		
Overall financial performance	0.83	41.91	0.69		
Entrepreneurial orientation (Covin & Slevin, 1989; Covin & Wales, 2012; Tang et al., 2008)				0.98	0.933
Risk-taking	0.91	56.96	0.82	0.88	0.870
Proclivity for high-risk/return projects	0.84	47.09	0.71		
Bold, wide-ranging acts are necessary	0.86	50.99	0.74		
Adopt a bold, aggressive posture	0.81	40.42	0.66		
Innovativeness	0.92	59.76	0.85	0.86	0.853
Strong emphasis on R&D and innovation	0.83	43.74	0.69		
Introduced many new lines of products/service	0.76	32.37	0.58		
Changes in product lines have been dramatic	0.84	46.43	0.71		
Proactiveness	0.99	88.77	0.99	0.84	0.833
Initiate competitive actions that others respond to	0.81	40.61	0.65		
Be the first to introduce new products/services	0.78	37.28	0.61		
Very competitive, undo-the-competitors posture	0.80	37.23	0.63		
Absorptive capacity (Cohen & Levinthal, 1990; Flatten et al., 2011; Wei, Lowry, & Seedorf, 2015)				0.92	0.915
Use external resources to obtain information	0.86	57.64	0.75		
Information communicated effectively intra-firm	0.83	46.91	0.69		
Able to structure and use collected knowledge	0.89	69.79	0.80		
Able to adopt and commercialize new technology	0.84	49.88	0.70		
Boundary-spanning (Camburn, Rowan, & Taylor, 2003; Luo, 2003; Peng & Luo, 2000)				0.88	0.866
Utilize personal ties with government officials	0.86	45.94	0.74		
Utilize personal ties with business partners	0.88	49.12	0.77		
Utilize personal ties with research institutes	0.75	29.71	0.56		

Our independent variable was EO. We adopted the conceptualization of EO as a sustained firm-level attribute represented by the common qualities of risk-taking, innovative, and proactive behaviours, following Miller (1983), Covin and Slevin (1989), and Tang et al. (2008). Covin and Wales (2012) suggest that a second-order reflective factor specification best matches this conceptualization of EO, and therefore this measurement model was used in this study. As shown in Table 2, all items were significantly associated with their corresponding first-order factor, and the first-order factors were all heavily loaded on the second-order factor (i.e., EO). The internal consistency reliabilities for first- and second-order factors were satisfactory as evidenced by Cronbach's alpha and coefficient H both above 0.8.

The first mediator, *ACAP*, is reflected in a firm's knowledge acquisition, assimilation, transformation, and exploitation processes (Cohen & Levinthal, 1990; Zahra & George, 2002). Flatten, Engelen, Zahra, and Brettela (2011) summarized the commonly used measurement items for these dimensions, and concluded that the dimensions are reflective of the overall construct of absorptive capacity. As the dimensions are manifestations of absorptive capacity, the subscales of the dimensions can be pooled to measure the overall construct (see Keh et al., 2007). Therefore, while we included the final scale items proposed by Flatten, Engelen, Zahra, and Brettela (2011) and Wei, Lowry, and Seedorf (2015), we used the summed scores of the four dimensions to measure ACAP for model parsimony. As shown in Table 2, the scores (based on acquisition, assimilation, transformation, and exploitation processes) of the four dimensions were all significantly associated with the overall construct, and the construct reliability was substantially high (α = 0.92; H = 0.92).

The second mediator, *BS*, was reflected in managerial activities to establish and utilize social ties with other firms, government officials (Luo, 2003; Park & Luo, 2001; Peng & Luo, 2000), and research institutes such as universities (Gao, Xu, & Yang, 2008). We followed Peng and Luo (2000) by using general questions regarding a firm's social ties with various actors in each category. A high level of networking capability to establish bridging social capital is reflected not only in a greater number of social ties, but also in the even spread of those ties across external stakeholder groups (Cao, Simsek, & Jansen, 2015). As such, when conceptualized as a capability, BS has been measured as a reflective construct in prior studies (e.g. Camburn, Rowan & Taylor, 2003; Luo, 2003). As with ACAP, we calculated summed scores for business, government, and research institute categories to measure the overall construct of managerial BS capability. Table 2 shows that the summed scores were significantly associated with the construct and the construct reliability was high (α = 0.87; H = 0.88).

The moderator was the development level of *market-supporting institutions* in the firm's operating environment. It was measured based on the NERI's marketization index of Chinese provinces (Fan et al., 2011), which captures province-level government intervention in the economy, private business development, commodity and factor market development, and development of market intermediaries, all normalized to a value ranging from 0 to 10, where a high value indicates a high level of market-supporting institutional development. This index was published annually from 1997 to 2009 (Fan et al., 2011), and from 2010 to 2012 (Wang, Yu & Fan, 2013). As our sample firms were established in different locations and at different times, we used the annual marketization index scores of each firm's provincial location, averaged across the years from the firm's founding to the time of the survey, to represent the institutional environment that the firm has been exposed to.

We also included several control variables using secondary data. *Firm size* was measured by the number of employees. *Firm age*, which is related to the experiential know-how of the firm, was measured by the number of years since the firm's establishment. *Industry* was measured by dummy variables representing the primary industry affiliation of the sample firms, which were IT, Bio & Medicine, New Energy, New Material, Environment, Electronic, Creative industry, and other industries. Firm *ownership* type was measured based on the ownership status of the firm, and this fell into one of four categories, which were state owned or controlled, private, foreign controlled, and others. Three dummy variables were applied to control the potential ownership effect. *Returnee-led* firm was another dummy variable which had a value of one if the founder or top manager of the firm was a Chinese returnee, namely a person who had studied and/or worked overseas before returning to China to head the focal firm. Lastly, as our sample firms were

Table 3
Descriptive statistics and correlation.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Performance	1.00														
2. Entreprenusrial orientation	0.86	1.00													
3. Absoprtive capacity	0.79	0.82	1.00												
4. Boundary-spanning	0.62	0.63	0.52	1.00											
5. Market-supporting institutions	0.54	0.42	0.43	0.39	1.00										
6. Size	0.06	−0.01	0.03	−0.15	−0.12	1.00									
7. Age	0.54	0.37	0.38	0.35	0.63	0.21	1.00								
8. Industry 1 (dummy)	−0.08	−0.07	0.01	−0.05	−0.04	−0.11	−0.01	1.00							
9. Industry 2 (dummy)	0.09	0.12	0.08	0.06	0.02	0.11	0.04	−0.21	1.00						
10. Industry 3 (dummy)	−0.03	−0.01	0.05	0.05	0.03	0.01	0.04	−0.22	−0.13	1.00					
11. Industry 4 (dummy)	0.02	−0.04	−0.05	−0.04	−0.01	0.02	0.02	−0.22	−0.13	−0.13	1.00				
12. Industry 5 (dummy)	−0.03	−0.04	−0.09	0.01	0.02	−0.03	0.05	−0.21	−0.13	−0.13	−0.13	1.00			
13. Industry 6 (dummy)	0.07	0.09	0.01	0.11	0.05	0.01	−0.05	−0.24	−0.15	−0.15	−0.15	−0.14	1.00		
14. Industry 7 (dummy)	−0.02	0.05	0.02	0.04	−0.14	−0.16	−0.23	−0.16	−0.10	−0.10	−0.10	−0.10	−0.11	1.00	
15. Ownweship 1 (dummy)	0.02	−0.09	−0.04	0.10	0.02	0.32	0.11	−0.13	0.13	0.10	0.02	0.03	−0.05	−0.09	1.00
16. Ownweship 2 (dummy)	−0.03	0.04	0.04	−0.07	0.03	−0.40	−0.06	0.18	−0.21	−0.02	−0.13	−0.03	0.09	0.05	−0.63
17. Ownweship 3 (dummy)	0.01	0.02	−0.03	−0.02	−0.07	0.21	−0.04	−0.10	0.13	−0.06	0.15	0.02	−0.06	0.02	−0.12
18. Returnee-led (dummy)	−0.13	−0.05	−0.03	−0.07	−0.05	−0.04	−0.14	0.12	−0.04	−0.09	0.16	−0.14	−0.02	0.03	−0.13
19. Cluster 1 (dummy)	0.52	0.42	0.36	0.48	0.80	−0.32	0.48	−0.01	0.01	0.05	−0.06	0.02	0.12	−0.13	0.00
20. Cluster 2 (dummy)	−0.12	−0.15	−0.08	−0.08	−0.06	0.10	−0.02	−0.01	−0.02	−0.01	−0.05	0.01	−0.07	0.08	0.13
21. Cluster 3 (dummy)	−0.11	−0.06	−0.01	−0.27	0.01	0.18	−0.18	0.00	−0.02	−0.08	0.06	−0.05	−0.05	0.04	−0.11
Mean	5.14	5.02	5.07	4.75	9.00	3.18	3.30	0.26	0.11	0.12	0.12	0.11	0.14	0.07	0.10
S.D.	0.87	0.82	0.86	0.93	1.15	1.56	0.94	0.44	0.32	0.32	0.32	0.31	0.35	0.25	0.31

	16	17	18	19	20	21
16. ownweship 2 (dummy)	1.00					
17. ownweship 3 (dummy)	−0.67	1.00				
18. Returnee-led (dummy)	−0.05	0.20	1.00			
19. Cluster 1 (dummy)	0.10	−0.14	−0.01	1.00		
20. Cluster 2 (dummy)	−0.10	0.02	−0.04	−0.32	1.00	
21. Cluster 3 (dummy)	−0.01	0.09	0.01	−0.37	−0.20	1.00
Mean	0.77	0.12	0.17	0.37	0.15	0.19
S.D.	0.42	0.32	0.38	0.48	0.36	0.39

n = 411, r's > |0.10| are significant at p < 0.05.

from four high-tech clusters, we controlled for potential cluster effects using three cluster dummy variables. Table 3 below shows the descriptive statistics and correlations of the variables.

4. Analyses and results

4.1. Analytical procedures

Anderson and Gerbing's (1988) comprehensive two-stage SEM analytical procedure was followed in the data analysis. The first stage was to assess our measurement model. Alternative measurement models were also tested to establish the discriminant validity of the constructs in the original measurement model, as well as to detect any potential common method bias. The second stage estimated a theoretical model for hypothesis testing. For all SEM models, we assessed and reported multiple model fit indicators as suggested by Hu and Bentler (1999).

4.2. Measurement model

We first checked the normality of the data. The results indicated that the values of skew and kurtosis were below the cut-off values of 3 and 8 respectively, which satisfied the normality requirement for our analyses. The measurement model included the main constructs and their corresponding measurement items as described above. The model test is essentially a CFA procedure, of which the results are reported in Table 2. As reported in Table 2, the constructs showed satisfactory convergence validity evidenced by high reliability H and Cronbach's alpha. Overall, the model had adequate fit as evidenced by an χ^2 ratio of 2.86, a CFI of 0.962, and a SRMR of 0.031, which were all in the desired ranges proposed by Hu and Bentler (1999).

For a further construct validity test, we compared our original measurement model (second-order reflective construct of EO) with multiple alternative measurement models. The first alternative was a model where EO was measured as a first-order reflective construct (see Covin & Wales, 2012). Compared to the original model, this alternative model gained three degrees of freedom, and an χ^2 increase of 89.073. This χ^2 difference is significant at the 0.001 level, which suggests a deteriorated model fit. Accordingly, the original second-order reflect construct of EO was confirmed to be the better measurement model for our dataset.

We also tested other alternative measurement models where the four latent constructs, namely performance, EO, ACAP, and BS, were merged two at a time. χ^2 difference tests suggested that introducing such constraints significantly worsened the model fit when compared to the original model. In other words, the four latent constructs should not be converged with one another, which suggests construct discriminant validity.

A final alternative measurement model was tested where all indicator items for all variables were modelled to measure a single factor. This single factor model is equivalent to Harman's one factor test which is often used to detect common method variance (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). As expected, the model returned the lowest model fit, where all model fit indicators were significantly outside the desired range. This result suggested that our data was not biased by any underlying common method factors. Considering our multi-source design, we also tested within-subject common method variance using the same procedure, and the results were largely consistent with the single factor mode. Overall, we did not detect common method bias in our data.

In order to further detect common method variance and to ensure reliability and validity, we tested two alternative measurement models. In the first alternative model, we included a method factor in the full

CFA with all items loading onto that factor as well as onto their assigned construct. The model had adequate fit as evidenced by an χ^2 ratio of 2.489, a CFI of 0.974, and a SRMR of 0.025. Another alternative model contained two respondent factors reflecting the multi-respondent nature of our survey data. Specifically, the items of EO and performance were loaded onto the first respondent factor and their assigned construct, while the items of ACAP and BS were loaded onto the second respondent factor and their assigned construct. This model also had adequate fit as evidenced by an χ^2 ratio of 2.702, a CFI of 0.970, and a SRMR of 0.027. Chi-square difference tests suggested that these alternative models did not improve significantly in terms of goodness-of-fit from the original measurement model. In conclusion, we found sufficient convergent and discriminant validity and no evidence of common method variance for our measurement model.

4.3. Structural model and hypotheses testing

We followed James and Brett's (1984) recommendation of using confirmatory techniques, such as SEM, to test causal mediation relationships. We simultaneously estimated the effects from EO to the mediators, and from the mediators to performance, in a structural model (see Fig. 1). This joint significance test approach, as examined by MacKinnon, Lockwood, Hoffman, West, and Sheets (2002) in their simulation study, offers the best balance of Type I error and statistical power among the alternative approaches of testing the mediating effect.

As shown in Fig. 1, we specified our model as a complete mediation model. James, Mulaik, and Brett (2006) suggest that when there is no existing theory suggesting an immediate effect of the causal variable on the outcome variable, its direct effect should be constrained to zero when testing mediation. Shrout and Bolger (2002) also concur that assuming a complete mediation model has important statistical and conceptual advantages as the complete mediation model is the most parsimonious mediation model, and parsimonious models are taken as the theoretical baselines in science as they are the easiest to reject. Lastly, constraining the direct effect to zero can also improve the statistical power of detecting any mediating effects.

The complete mediation was supported in our χ^2 difference test by comparing the model fits between a constrained model (complete mediation) and a non-constrained model (EO-performance freely estimated). Our original model (complete mediation) returned an χ^2 ratio of 2.55, a CFI of 0.951, and a SRMR of 0.043 which suggests a good model fit. The non-constrained model had an χ^2 difference of 2.132 from the constrained model, with one degree of freedom, which was statistically insignificant. These results suggest that the two models were not significantly different in terms of goodness-of-fit to data, and therefore the more parsimonious model (i.e., the complete mediation model) should be adopted. These results suggest that the effect of EO on performance is fully mediated through the mechanisms of ACAP and BS.

Supports to individual mediating hypotheses are inferred from the parameter estimates related to each mediator. Based on the joint significance test approach (MacKinnon, Lockwood, Hoffman, West, & Sheets, 2002), a mediation is supported if the causal path from the independent variable to the mediator, and that from the mediator to the dependent variable, are jointly non-zero. As shown in our results (Fig. 1), EO had a positive and significant effect on ACAP (b = 0.833, $p < 0.001$), and ACAP had a positive and significant effect on performance (b = 0.630, $p < 0.001$). These results suggest that ACAP positively mediates the positive relationship between EO and performance. We estimated the size of the indirect effect from EO to performance through ACAP using the bootstrap technique in SEM (Shrout & Bolger, 2002). The estimated indirect effect was 0.591 ($p < 0.001$) with a 95% confidence interval from 0.423 to 0.716, and this accounted for 77.6% of the total effect of EO on performance. Overall, our hypothesis 1 was supported. For hypothesis 2, the relationship between EO and BS was positive and significant (b = 0.060, $p < 0.001$), and BS had a positive and significant association with performance (b = 0.160, $p < 0.05$). The results of the bootstrap analysis showed that the estimate of the indirect effect of EO on performance through BS was 0.171 ($p < 0.01$) with a 95% confidence interval from 0.096 to 0.334, and that this accounted for 22.4% of the total effect of EO on performance. Our hypothesis 2 was therefore also supported.

Several results of the control variables are also worth noting. Foreign ownership control was negatively related to ACAP (b = − 0.330, $p < 0.05$), but positively related to firm performance (b = 0.361, $p < 0.05$). This suggests that when operating in China, foreign owned firms have competitive advantages originating from internal knowledge sources, rather than through external knowledge searches. Firm size was negatively related to BS (b = − 0.113, $p < 0.01$), but positively related to performance (b = 0.047, $p < 0.05$). This may be due to the fact that larger firms are less resource-dependent on external networks for competitive advantages. Firm age was positively related to both BS (b = 0.153, $p < 0.01$) and performance (b = 0.108, $p < 0.01$). Long-established firms are in a

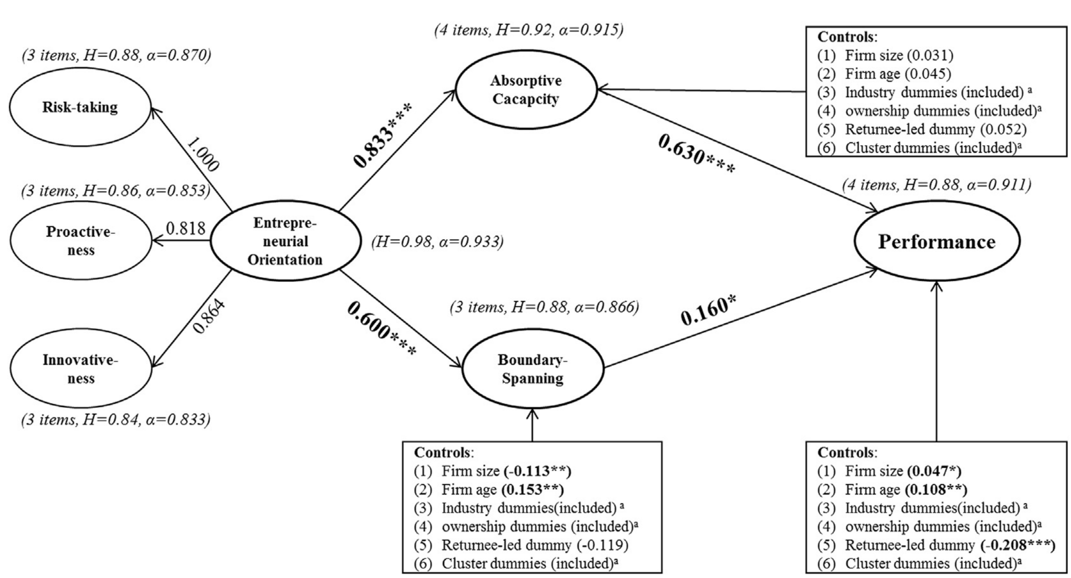


Fig. 1. SEM analysis for mediating hypothesis test.
Note: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; a: dummy variables' results are excluded for brevity.

Table 4
Multi-group SEM analysis for moderating hypothesis test.

	Model 1: Unconstrained	Model 2: Fully constrained	Model 3: ACAP path constrained	Model 4: BS path constrained	Model 5: EO-ACAP constrained	Model 6: ACAP-Performance constrained	Model 7: EO-BS Constrained	Model 8: BS-Performance Constrained
Low market								
EO → ACAP	0.807**,***	0.833***	0.847***	0.800***	0.847***	0.807***	0.791***	0.816***
ACAP → PER	0.380***	0.630***	0.667***	0.492***	0.382***	0.666***	0.399***	0.478***
EO → BS	0.807***	0.600***	0.814***	0.655***	0.814***	0.804***	0.657***	0.808***
BS → PER	0.332***	0.160*	0.098*	0.222***	0.326***	0.109***	0.328***	0.225***
High market								
EO → ACAP	0.869***	0.833***	0.847***	0.876***	0.847***	0.875***	0.881***	0.865***
ACAP → PER	0.721***	0.630***	0.667***	0.694***	0.724***	0.666***	0.714***	0.702***
EO → BS	0.493***	0.600***	0.496***	0.655***	0.491***	0.499***	0.657***	0.493***
BS → PER	0.187***	0.160*	0.215***	0.222***	0.190***	0.212***	0.184***	0.225***
Chi-square (χ^2)	1792.257	1824.530	1806.384	1809.146	1793.100	1805.279	1806.259	1795.159
df	760	764	762	762	761	761	761	761
$\Delta\ \chi^2$		32.273	14.127	16.889	0.843	13.022	14.002	2.902
Δ *df*		4	2	2	1	1	1	1
p-value		0.0000	0.0009	0.0002	0.3585	0.0003	0.0002	0.0885

Note: Control variables are excluded from this table for brevity.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

better position to penetrate network barriers and to gain experience-based competitive advantages. Lastly, the returnee-led dummy had a negative effect on performance (b = −0.208, $p < 0.001$), which suggests that firms founded by overseas returnees under-perform those with local founders. Lack of local market knowledge may contribute to this negative effect.

The moderation hypotheses (H3a and b) were tested using multi-group SEM analysis procedures following Song, Droge, Hanvanich, and Calantone (2005). The aim of these tests was to reveal whether the mediated pathways should be allowed to vary in strength between sub-groups divided by the value of the moderator. The results, as shown in Table 4, compare the goodness-of-fit of a series of alternative structure models.

Model 1 is the baseline model where only the control variables and factor loadings were constrained, while the hypothesized mediating pathways were allowed to vary between subsample groups of firms operating in high versus low level marketization environments, split by the mean value of market-supporting institutions. All subsequent models in Table 4 impose certain constraints on this baseline model against the hypothesized moderating effects. Specifically, model 2 constrained all four paths involved in the mediating pathways (i.e., EO-ACAP, EO-BS, ACAP-performance, and BS-performance); model 3 constrained the mediating pathway through ACAP (i.e., EO-ACAP, ACAP-performance); model 4 constrained the mediating pathway through BS (i.e., EO-BS, BS-performance). All of these models returned significant results from chi-square difference tests against the baseline model. This suggests that the baseline model fits the data better when the mediating pathways are allowed to vary in strength depending on the development of market-supporting institutions in the operating environments of firms. In contrast, constraining the mediating pathways despite the different levels of the moderator significantly deteriorated model fit. These results provide general support for the moderated mediation relationships.

Specific to hypothesis 3a, the results of model 1 show that the EO-ACAP (0.869 > 0.807) and ACAP-performance (0.721 > 0.380) coefficients were both larger within the high market-supporting institution group than within the low market-supporting institution group. The difference in strength for the EO-ACAP path was, however, not statistically significant, as shown by the model chi-square difference test of model 5 ($\Delta\chi^2_{(1)} = 0.843$, $p > 0.05$), as constraining the EO-ACAP path between the groups did not significantly worsen the model fit. The difference in strength for the ACAP-performance path was statistically significant given the model chi-square difference test of model 6 ($\Delta\chi^2_{(1)} = 13.022$, $p < 0.001$). Regarding hypothesis 3b, the results of model 1 show that the EO-BS (0.493 < 0.807) and BS-performance (0.187 < 0.332) coefficients were both smaller within the high market-supporting institution group than within the low market-supporting institution group. The difference in strength for the EO-BS path was statistically significant as shown by the model chi-square difference test of model 7 ($\Delta\chi^2_{(1)} = 14.002$, $p < 0.001$), whereas the difference in strength for the BS-performance path was not statistically significant given the model chi-square difference test of model 8 ($\Delta\chi^2_{(1)} = 2.902$, $p > 0.05$). Overall, these results provide support to both hypotheses 3a and 3b by showing that the mediated pathway between EO and performance via ACAP is stronger, while that via BS is weaker, when market-supporting institution is at a higher than at a lower level. The detailed results from Table 4 also enabled us to pinpoint the moderation effect for both mediating pathways. The moderation was on the second stage for the ACAP mediating pathway (i.e., ACAP-performance) and on the first stage for the BS mediating pathway (i.e., EO-BS).

We also followed Kunze, Jong, and Bruch (2016) when comparing the mediated indirect effects between subsample groups with low versus high levels of market-supporting institutions. Using bootstrap techniques of SEM, we found that the indirect effect of EO on performance via ACAP was not statistically significant in the low market-supporting institution group (95% confidence interval from −0.085 to 0.318), but this indirect effect was higher and significant in the high market-supporting institution group (95% confidence interval from 0.436 to 0.844). The indirect effect of EO on performance via BS was higher in the low market-supporting institution group (95% confidence interval from 0.318 to 0.844), than in the high market-supporting institution group (95% confidence interval from 0.033 to 0.235). Given that there was no overlap of the confidence intervals associated with the respective indirect effects between the groups, we found further support for our hypotheses (H3a/b).

To further evaluate these findings, we also estimated a full structural model of the moderated mediation relationships (see Appendix B), based on the integration algorithm of Mplus 7.0. As shown in Appendix B, the estimates of the interaction terms were consistent with the above multi-group SEM procedure findings. Specifically, market-supporting institutions positively moderated the ACAP-performance relationship (b = 0.089, $p < 0.05$) but not the EO-ACAP relationship, and they negatively moderated the EO-BS relationship (b = −0.163,

$p < 0.001$) but not the BS-performance relationship.

4.4. Additional analyses

We conducted post-hoc analyses to rule out potential reverse causalities (i.e. performance may support high EO and the development of ACAP and BS). Although our theoretical development indicates causality in our mediating hypotheses, empirically we were limited based on using cross-sectional data to fully support it. Therefore, following the suggestion of Mesmer-Magnus, Asencio, Seely, and DeChurch (2015), we explored models with alternative causal linkages among the key variables and evaluated their model fit. In the first model, we treated performance as an independent variable, ACAP and BS as mediators, and EO as a dependent variable. In the second model, we held performance as the independent variable, while modelling EO as a mediator and ACAP and BS as two dependent variables. In the third model, we tested the effect of performance on ACAP, BS, and EO simultaneously. In the fourth model, we explored the possibility that there exists a reciprocal relationship between EO and BS. None of these models returned sufficient model fit, indicated by the χ^2/df ratios exceeding the acceptable threshold of 3.0., as well as by other model fit indicators (see Appendix C for details). Accordingly, the alternative causal linkages were not supported by our data.

5. Discussion

5.1. Research contributions

Explicating the mediating mechanisms underlying the EO-performance relationship remains an important yet underexplored research area in the EO literature (Lumpkin & Dess, 1996; Wales et al., 2013). We have attempted to address this knowledge gap in this study by arguing that EO as a strategic posture influences firm performance outcomes through the mechanisms of knowledge-based and network-based dynamic capability development, the effectiveness of which is contingent on market institutional development. The findings of this study contribute to the EO literature by showing that the performance effect of EO is fully mediated by the dynamic capabilities that support entrepreneurial behaviours, and that the mediating effect of the knowledge-based and network-based dynamic capabilities alternate depending on the institutional environment the entrepreneurial firm operates in. This full mediation model helps to address the drawback of conceptualizing EO as a dispositional construct, regarding the distinctiveness of EO from other intangible firm-level attributes such as organizational culture and climate (Covin & Lumpkin, 2011). Our findings suggest that EO serves as an initiator of organizational processes aimed at developing dynamic capabilities to support entrepreneurial behaviours. It is therefore possible to conceptually and empirically distinguish EO from other organizational intangibles based on its mediating capabilities. For example, in an emerging market context EO can be more specifically defined and modelled as a firm-level strategic posture to engage in ACAP and BS capability development to support risk-taking, innovative, and proactive behaviours.

Moreover, our findings also contribute to the EO literature by introducing and empirically testing the moderating effects of market-supporting institutions on the two mediating pathways linking EO with firm performance. Revealing the institutional contingencies of the EO effect answers the calls of scholars such as Wales et al.'s (2013) contention that the EO-performance relationship is contingent on environmental factors, and Yang and Su's (2014) insights that industrial marketing research requires a deep understanding of the complex nature of institutions in the setting of business markets. Furthermore, the empirical context of this study allowed us to evaluate institutional contingencies at the sub-national level. This effort advances our understanding of the institutional complexity that exists in sizeable emerging economies such as China, where significant within-country variances exist across regions (Ahlstrom & Bruton, 2010; Xia, Boal, & Delios, 2009). Specifically, in the context of regional differences in China, our findings suggest that entrepreneurial firms are better positioned to succeed if they develop ACAP in provinces with more developed market-supporting institutions, while they have a higher chance of success in provinces with less developed market-supporting institutions by focussing on bridging and leveraging networks.

Furthermore, this study contributes to EO as “a needed construct” (Covin & Lumpkin, 2011: 855) by extending its application to the emerging market context. Scholars have called for more entrepreneurship research on business venturing in institutional and competitive environments that are different from mature economies (Hoskisson, Covin, Volberda, & Johnson, 2011; Tang et al., 2008; Wales et al., 2013). Firms within emerging markets face different external environment and associated challenges in their pursuit of entrepreneurship than those faced by firms within developed countries. Accordingly, while EO as a strategic posture is a universal concept, the mechanisms through which EO impacts performance are contextually specific rather than universal, as a sufficient strategy-environment fit is essential for business success (Zajac, Kraatz, & Bresser, 2000). Prior studies have predominantly focused on organizational learning as the sole mediating mechanism of the EO-performance relationship within a developed country context (e.g. Li, Huang, & Tsai, 2009; Wang, 2008). Our findings suggest that in the context of an emerging market, entrepreneurial firms can alternate between two channels of capability development to realize the performance effect of EO. In other words, we have shown that firms in emerging economies can employ organizational leaning (Flores, Zheng, Rau, & Thomas, 2012) and nonmarket strategies (Hitt, Lee, & Yucel, 2002) to attain entrepreneurial profits.

5.2. Managerial implications

This study presents two main practical implications for managers of emerging market firms. An implication directly derived from our findings is that firms need to actively engage in capability development in order to realize the potential performance benefits of EO. While the possession of EO is a necessary prerequisite for the realization of entrepreneurial profits, EO by itself is not sufficient. Managers need to initiate, manage, and coordinate organizational processes to develop dynamic capabilities such as ACAP and BS to transform EO into superior performance. Specifically, in the context of an emerging market, managers should evaluate the scope and depth of their social networks to determine if they could be utilized to support risky, proactive, and innovative activities. Firms without strong social ties may consider acquiring such capabilities by recruiting BS managers (cf. Zhang, Wu, & Henke, 2015). For example, one of the potential contributions of returnee managers is their established business networks with foreign firms, which can be leveraged for know-how learning and international expansion (Tan & Meyer, 2010). Firms can also outsource the network building process to institutional entrepreneurs, such as individuals who are well endowed with social capital based on kinship or privileged social status. Managers should also emphasize the accumulation of experiential knowledge to support effective organizational learning and the development of new resource bundles. Experiential knowledge can foster absorptive capacity, which in turn increases the likelihood of the success of future innovations.

A second implication is that firms need to manage the environmental complexity of the underlying mechanisms supporting the positive performance effect of EO. Firms may engage in multiple businesses simultaneously in various locations, and managers should be aware of the development of market-supporting institutions in these locations. For example, in our empirical context, BS serves as an effective channel for the transformation of EO to superior performance, and the mediating effect is strengthened by weaker market institutions. Accordingly, managers may devote more attention and other organizational resources to the BS channel when doing business in a location that features institutional voids. However, the effectiveness of the two channels may evolve overtime as the external business environment and the internal capabilities of firms develop. Therefore, managers need to constantly evaluate the strategic fit of their EO-induced capability development and make necessary adjustments accordingly. Managerial agency therefore plays a critical role in maximizing the positive performance effect of EO.

5.3. Limitations and future research directions

This study has two main limitations. First, the empirical test for this study is based on a single country context. Although the Chinese context offers a number of advantages in testing our model, there may be country-specific effects that are potentially influential. Future research could replicate our study in a different empirical setting to validate our results. However, a more fruitful direction for future research, as suggested above, is to extend our study, both theoretically and empirically, to a cross-country context which allows for the examination of contextual effects.

Second, although we have used a multi-source design to minimize common method variance, the inherent disadvantages of the survey design could not be completely eliminated. For example, although the self-rated performance measure has been widely used in prior studies, potential bias does exist for this method. There is the possibility that this bias could be addressed by using secondary data on a firm's financial performance. By using regularly published secondary data, future research could also adopt a longitudinal design to investigate the long-term effects of EO on firm performance outcomes.

Future research could extend the current study in several directions. First, the contextual variances in the EO-performance mediating mechanisms could be explicitly examined in studies with cross-country and/or longitudinal design. The relative effectiveness and interaction of the multiple mediating mechanisms underlying the EO-performance relationship may vary across different contexts. The current study focuses on an emerging market context using China as the empirical setting, however, research has suggested that there are institutional and economic dissimilarities among emerging economies (Xu & Meyer, 2013). Future research using a multi-country and/or longitudinal design could reveal the important contextual factors that moderate the mediating relationships identified in the current study, and thus contribute to a more holistic understanding of the performance effect of EO.

Second, while we have discussed how the relative effectiveness of ACAP and BS channels varies depending on the level of development of market-supporting institutions, it is likely that firms will pursue both channels, albeit to varying degrees, as long as their institutional environments do not sit on either extremity of the spectrum of institutional development. Therefore, tension is likely to exist between ACAP and BS channels as they compete for managerial cognitive resources. This tension is likely to be managed differently across firms given the role of managerial intentionality (Hutzschenreuter, Pedersen, & Volberda, 2007). Future research could study the role of managerial leadership and how it can influence the level of tension tolerance in organizations with seemingly opposite strategic orientations (e.g. exploitation vs. exploration) (Raisch & Birkinshaw, 2008; Vera & Crossan, 2004). Broadly speaking, firms alternate between the two mediated channels of pursuing entrepreneurial success, not only depending on the external environment, but also due to being influenced by internal conditions. Future research could explore a variety of organizational contingencies for the mediated EO-performance relationship.

Last but not least, future research could also explore the role of EO in a larger firm context. The generative mechanisms underlying the EO-performance relationship for larger firms may differ from those for SMEs. Also, social network analysis techniques could be combined with SEM to gain a greater understanding of the effects of network ties and boundary spanning on the attitudes, behaviours, and performance of firms.

Acknowledgement

We thank Professor Adam Lindgreen and three anonymous reviewers and for their constructive comments on earlier versions of this paper. The first author thanks research support from the National Natural Science Foundation of China (71472038).

Appendix A. In-depth qualitative case studies

Supplemental Qualitative Research Design

In addition to the deductive hypothesis development process, we conducted qualitative multiple case studies to supplement more contextualized information and to elaborate on the two mediating mechanisms and their regional institutional contingencies. In line with Pahnke, Katila, and Eisenhardt (2015), the role of this qualitative information was not to corroborate the hypothesized mechanisms and contingencies, but to supplement and enhance hypothesis development. We collected primary qualitative data from in-depth interviews. Four case firms were selected based on their representativeness (see Table below). We triangulated these interviews by talking to multiple funding partners and managerial team members, supplemented by corporate archival data, such as corporate reports and meeting minutes. All interviews with the four case firms lasted between 3 and 5 h, and occurred between June 2016 and April 2017. Following Pahnke, Katila, and Eisenhardt (2015), we used two interview protocols which included chronological questions about their venture history, strategic orientation, capability development, and performance outcomes. The interviews were semi-structured, and we guided the interviewees to elaborate on the roles of knowledge- and network-based capabilities in achieving their entrepreneurial success.

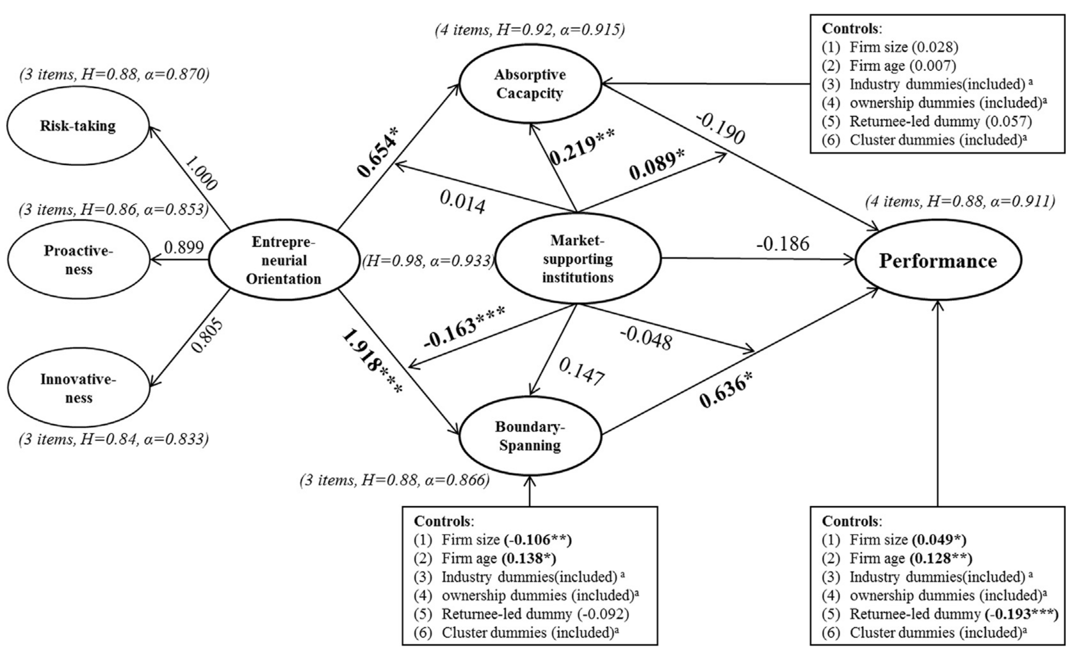
Profiles of case companies and interview summaries

Case firm	Shifang	Jinru	Zhonghe Tech.	BLB biology
Company details				
Ownership	PE[a]	PE	PE	PE
Year of incorporation	1994	2000	2011	1997
Major products/ focus	Automation engineering works/digital frequency converter	Renewable energy/biological power generation	Online platform technology	Corn-based health and food ingredients
Head office	Shanxi province	Beijing	Beijing	Shandong province

Corporate size	Small	Small	Small	Medium
Entrepreneurial type[b]	A Specialist (who is keen on learning new technology; has expertise in his field; prefers technical work to administrative tasks)	An Opportunist (who has broader experience, and is good at analysing the best opportunity and utilising all resources around him to take risk and invest)	A Builder (who is a fast decision maker, and good at organising teams and building a good working platform for employees)	An Innovator (who is actively inventing, and creating breakthrough ideas; and believes that innovative technology and refinery process generate competitive advantages)
Revenue in 2016 (RMB Million)	20	100	250	1377
Interview details				
Interviewees	Founding CEO	Founding chairman	Founding CEO	Founding general manager
Qualitative triangulation	Interview co-founder/CFO, and chief engineer for comparison; archive data	Interview co-founder/former general manager, and CFO for comparison	Interview co-founder/COO for comparison	Archive data; interview co-founder/former CFO for comparison
Interview methods	F2F	F2F and Telephone interviewing	F2F	F2F
Interview time	Sept., 2016; Apr., 2017	Jun. & Sept., 2016; Apr., 2017	Sept., 2016	Nov., 2016
Interview location	Beijing, China/Melbourne, Australia	Beijing/Shanghai, China	Beijing, China	Wuxi, China
Total interview time	5 h	5 h	3 h	3 h

[a] PE = Private enterprise; CEO = Chief Executive Officer; CFO = Chief Financial Officer; COO = Chief Operation Officer; F2F = face to face interview; Sept. = September; Jun. = June; Apr. = April; Nov. = November; EO = Entrepreneurial orientation; hrs = hours.
[b] Entrepreneurial type refers to the entrepreneurship typology in the book entitled "Entrepreneurial DNA" by Joe Abraham, 2011, McGraw-Hill: New York.

Appendix B. Additional test of moderating hypotheses using SEM



Note: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; a: dummy variables' results are excluded for brevity.

Appendix C. Assessing alternative causal structures

	Model		χ^2	*df*	χ^2/df	CFI	RMSEA	SRMR
IV: performance	M: ACAP, BS	DV: EO	612.08	146	4.192	0.931	0.088	0.044
IV: performance	M: EO	DV: ACAP, BS	504.60	146	3.456	0.947	0.077	0.039
IV: performance		DV: ACAP, BS, EO	525.43	146	3.599	0.944	0.080	0.040
IV: BS	M: EO	DV: performance	330.47	85	3.888	0.951	0.084	0.040

Note: IV: independent variable; M: mediator; DV: dependent variable.

References

Ahlstrom, D., & Bruton, G. D. (2010). Rapid institutional shifts and the co-evolution of entrepreneurial firms in transition economies. *Entrepreneurship Theory and Practice, 34*(3), 531–554.

Aldrich, H., & Herker, D. (1977). Boundary spanning roles and organization structure. *Academy of Management Review, 2*(2), 217–230.

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin, 103*(3), 411–423.

Banalieva, E. R., Eddleston, K. A., & Zellweger, T. M. (2015). When do family firms have an advantage in transitioning economies? Toward a dynamic institution-based view. *Strategic Management Journal, 36*(9), 1358–1377.

Buckley, P. J., Clegg, L. J., Cross, A. R., Liu, X., Voss, H., & Zheng, P. (2007). The determinants of Chinese outward foreign direct investment. *Journal of International Business Studies, 38*(4), 499–518.

Camburn, E., Rowan, B., & Taylor, J. E. (2003). Distributed leadership in schools: The case of elementary schools adopting comprehensive school reform models. *Educational Evaluation and Policy Analysis, 25*(4), 347–373.

Cao, Q., Simsek, Z., & Jansen, J. J. (2015). CEO social capital and entrepreneurial orientation of the firm: Bonding and bridging effects. *Journal of Management, 41*(7), 1957–1981.

Carroll, G. R., & Teo, A. C. (1996). On the social network of managers. *Academy of Management Journal, 39*(2), 421–440.

Chang, S. J., Chung, C. N., & Mahmood, I. P. (2006). When and how does business group affiliation promote firm innovation? A tale of two emerging economies. *Organization Science, 17*(5), 637–656.

Chen, Y. S., Lin, M. J. J., & Chang, C. H. (2009). The positive effects of relationship learning and absorptive capacity on innovation performance and competitive advantage in industrial markets. *Industrial Marketing Management, 38*(2), 152–158.

Chen, Y. C., Li, P. C., & Evans, K. R. (2012). Effects of interaction and entrepreneurial orientation on organizational performance: Insights into market driven and market driving. *Industrial Marketing Management, 41*(6), 1019–1034.

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly, 35*(1), 128–152.

Covin, J. G., & Lumpkin, G. T. (2011). Entrepreneurial orientation theory and research: Reflections on a needed construct. *Entrepreneurship Theory and Practice, 35*(5), 855–872.

Covin, J. G., & Miller, D. (2014). International entrepreneurial orientation: Conceptual considerations, research themes, measurement issues, and future research directions. *Entrepreneurship Theory and Practice, 38*(1), 11–44.

Covin, J. G., & Slevin, D. P. (1989). Strategic management of small firms in hostile and benign environments. *Strategic Management Journal, 10*(1), 75–87.

Covin, J. G., & Wales, W. J. (2012). The measurement of entrepreneurial orientation. *Entrepreneurship Theory and Practice, 36*(4), 677–702.

Dess, G. G., Pinkham, B. C., & Yang, H. (2011). Entrepreneurial orientation: Assessing the construct's validity and addressing some of its implications for research in the areas of family business and organizational learning. *Entrepreneurship Theory and Practice, 35*(5), 1077–1090.

Dollinger, M. J. (1984). Environmental boundary spanning and information processing effects on organizational performance. *Academy of Management Journal, 27*(2), 351–368.

Fan, G., Wang, X., & Zhu, H. (2011). *NERI index of marketization of China's provinces.* Beijing: Economic Science Press (in Chinese).

Flatten, T. C., Engelen, A., Zahra, S. A., & Brettela, M. (2011). A measure of absorptive capacity: Scale development and validation. *European Management Journal, 29*(2), 98–116.

Flores, L. G., Zheng, W., Rau, D., & Thomas, C. H. (2012). Organizational learning: Subprocess identification, construct validation, and an empirical test of cultural antecedents. *Journal of Management, 38*(2), 640–667.

Gao, S., Xu, K., & Yang, J. (2008). Managerial ties, absorptive capacity, and innovation. *Asia Pacific Journal of Management, 25*(3), 395–412.

Geletkanycz, M. A., & Hambrick, D. C. (1997). The external ties of top executives: Implications for strategic choice and performance. *Administrative Science Quarterly, 42*(4), 654–681.

Hitt, M. A., Lee, H. U., & Yucel, E. (2002). The importance of social capital to the management of multinational enterprises: Relational networks among Asian and Western firms. *Asia Pacific Journal of Management, 19*(2), 353–366.

Holburn, G. L., & Bergh, R. G. V. (2008). Making friends in hostile environments: Political strategy in regulated industries. *Academy of Management Review, 33*(2), 521–540.

Hoskisson, R. E., Eden, L., Lau, C. M., & Wright, M. (2000). Strategy in emerging economies. *Academy of Management Journal, 43*(3), 249–267.

Hoskisson, R. E., Covin, J., Volberda, H. W., & Johnson, R. A. (2011). Revitalizing entrepreneurship: The search for new research opportunities. *Journal of Management Studies, 48*(6), 1141–1168.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling, 6*(1), 1–55.

Hutzschenreuter, T., Pedersen, T., & Volberda, H. W. (2007). The role of path dependency and managerial intentionality: A perspective on international business research. *Journal of International Business Studies, 38*(7), 1055–1068.

James, L. R., & Brett, J. M. (1984). Mediators, moderators, and test for mediation. *Journal of Applied Psychology, 69*(2), 307–321.

James, L. R., Mulaik, S. A., & Brett, J. M. (2006). A tale of two methods. *Organizational Research Methods, 9*(2), 233–244.

Keh, H. T., Nguyen, T. T. M., & Ng, H. P. (2007). The effects of entrepreneurial orientation and marketing information on the performance of SMEs. *Journal of Business Venturing, 22*(4), 592–611.

Khanna, T., & Palepu, K. G. (1997). Why focused strategies may be wrong for emerging markets. *Harvard Business Review, 75*(4), 41–48.

Khanna, T., & Palepu, K. G. (2010). *Winning strategies for emerging economies.* Cambridge, MA: Harvard Business School Press.

Kollmann, T., & Stöckmann, C. (2014). Filling the entrepreneurial orientation–performance gap: The mediating effects of exploratory and exploitative innovations. *Entrepreneurship Theory and Practice, 38*(5), 1001–1026.

Kunze, F., De Jong, S. B., & Bruch, H. (2016). Consequences of collective-focused leadership and differentiated individual-focused leadership: Development and testing of an organizational-level model. *Journal of Management, 42*(4), 886–914.

Li, Y., Huang, J., & Tsai, M. (2009). Entrepreneurial orientation and firm performance: The role of knowledge creation process. *Industrial Marketing Management, 38*(4), 440–449.

Lisboa, A., Skarmeas, D., & Lages, C. (2011). Entrepreneurial orientation, exploitative and explorative capabilities, and performance outcomes in export markets: A resource-based approach. *Industrial Marketing Management, 40*(8), 1274–1284.

Lumpkin, G. T., & Dess, G. G. (1996). Clarifying the entrepreneurial orientation construct and linking it to performance. *Academy of Management Review, 21*(1), 135–172.

Luo, Y. D. (2003). Industrial dynamics and managerial networking in an emerging market: The case of China. *Strategic Management Journal, 24*(13), 1315–1327.

Luo, Y. D., Sun, J. Y., & Wang, S. L. (2011). Emerging economy copycats: Capability, environment, and strategy. *Academy of Management Perspectives, 25*(2), 37–56.

MacKinnon, D. P., Lockwood, C. M., Hoffman, J. M., West, S. G., & Sheets, V. (2002). A comparison of methods to test mediation and other intervening variable effects. *Psychological Methods, 7*(1), 83–104.

Mahon, J. F., & Murray, E. A. (1981). Strategic planning for regulated companies. *Strategic Management Journal, 2*(3), 251–262.

McDougall, P. P., Covin, J. G., Robinson, R. B., & Herron, L. (1994). The effects of industry growth and strategic breadth on new venture performance and strategy content. *Strategic Management Journal, 15*(7), 537–554.

Mesmer-Magnus, J. R., Asencio, R., Seely, P. W., & DeChurch, L. A. (2015). How organizational identity affects team functioning: The identity instrumentality hypothesis. *Journal of Management.*. http://dx.doi.org/10.1177/0149206315614370.

Miller, D. (1983). The correlates of entrepreneurship in three types of firms. *Management Science, 29*(7), 770–791.

Miller, D., & Friesen, P. H. (1978). Archetypes of strategy formulation. *Management Science, 24*(9), 921–933.

North, D. C. (1990). *Institutions, institutional change, and economic performance.* Cambridge, MA: Harvard University Press.

Pahnke, E. C., Katila, R., & Eisenhardt, K. M. (2015). Who takes you to the dance? How partners' institutional logics influence innovation in young firms. *Administrative Science Quarterly, 60*(4), 596–633.

Park, S. H., & Luo, Y. (2001). Guanxi and organizational dynamics: Organizational networking in Chinese firms. *Strategic Management Journal, 22*(5), 455–477.

Peng, M. W. (2003). Institutional transitions and strategic choices. *Academy of Management Review, 28*(2), 275–296.

Peng, M. W., & Heath, P. S. (1996). The growth of the firm in planned economies in transition: Institutions, organizations, and strategic choice. *Academy of Management Review, 21*(2), 492–528.

Peng, M. W., & Luo, Y. (2000). Managerial ties and firm performance in a transition economy: The nature of a micro–macro link. *Academy of Management Journal, 43*(3), 486–501.

Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903.
Raisch, S., & Birkinshaw, J. (2008). Organizational ambidexterity: Antecedents, outcomes, and moderators. *Journal of Management, 34*(3), 375–409.
Shane, S., & Venkataraman, S. (2000). The promise of entrepreneurship as a field of research. *Academy of Management Review, 25*(1), 217–226.
Shi, W. L., Sun, S. L., & Peng, M. W. (2012). Sub-national institutional contingencies, network positions, and IJV partner selection. *Journal of Management Studies, 49*(7), 1221–1245.
Shrout, P. E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods, 7*(4), 422–445.
Song, M., Droge, C., Hanvanich, S., & Calantone, R. (2005). Marketing and technology resource complementarity: An analysis of their interaction effect in two environmental contexts. *Strategic Management Journal, 26*(3), 259–276.
Tan, D. C., & Meyer, K. E. (2010). Business groups' outward FDI: A managerial resources perspective. *Journal of International Management, 16*(2), 154–164.
Tang, J., Tang, Z., Marino, L. D., Zhang, Y., & Li, Q. (2008). Exploring an inverted U-shape relationship between entrepreneurial orientation and performance in Chinese ventures. *Entrepreneurship Theory and Practice, 32*(1), 219–239.
Tracey, P., & Phillips, N. (2011). Entrepreneurship in emerging markets: Strategies for new venture creation in uncertain institutional contexts. *Management International Review, 51*(1), 23–39.
Tsai, W. (2001). Knowledge transfer in intraorganizational networks: Effects of network position and absorptive capacity on business unit innovation and performance. *Academy of Management Journal, 44*(5), 996–1004.
Ulaga, W., & Sharma, A. (2001). Complex and strategic decision making in organizations: Implications for personal selling and sales management. *Industrial Marketing Management, 30*(5), 427–440.
Vera, D., & Crossan, M. (2004). Strategic leadership and organizational learning. *Academy of Management Review, 29*(2), 222–240.
Wales, W. J., Gupta, V. K., & Mousa, F. T. (2013). Empirical research on entrepreneurial orientation: An assessment and suggestions for future research. *International Small Business Journal, 31*(4), 357–383.
Wang, C. L. (2008). Entrepreneurial orientation, learning orientation, and firm performance. *Entrepreneurship Theory and Practice, 32*(4), 635–657.
Wang, Q., Wong, T. J., & Xia, L. J. (2008). State ownership, the institutional environment, and auditor choice: Evidence from China. *Journal of Accounting & Economics, 46*(1), 112–134.
Wang, X., Yu, J., & Fan, G. (2013). *Provincial business environment index in China 2013.* Beijing, China: CITIC Press.
Wei, J., Lowry, P. B., & Seedorf, S. (2015). The assimilation of RFID technology by Chinese companies: A technology diffusion perspective. *Information Management, 52*(6), 628–642.
Wiklund, J., & Shepherd, D. (2003). Knowledge-based resources, entrepreneurial orientation, and the performance of small and medium-sized businesses. *Strategic Management Journal, 24*(13), 1307–1314.
Williamson, O. E. (1992). Markets, hierarchies, and the modern corporation: An unfolding perspective. *Journal of Economic Behavior & Organization, 17*(3), 335–352.
Xia, J., Boal, K., & Delios, A. (2009). When experience meets national institutional environmental change: Foreign entry attempts of U.S. firms in the central and eastern European region. *Strategic Management Journal, 30*(12), 1286–1309.
Xin, K. R., & Pearce, J. L. (1996). Guanxi: Good connections as substitutes for institutional support. *Academy of Management Journal, 39*(6), 1641–1658.
Xu, D., & Meyer, K. E. (2013). Linking theory and context: 'Strategy research in emerging economies' after Wright et al. (2005). *Journal of Management Studies, 50*(7), 1322–1346.
Yang, M., & Gabrielsson, P. (2017). Entrepreneurial marketing of international high-tech business-to-business new ventures: A decision-making process perspective. *Industrial Marketing Managemen.*. https://doi.org/10.1016/j.indmarman.2017.01.007.
Yang, Z., & Su, C. (2014). Institutional theory in business marketing: A conceptual framework and future directions. *Industrial Marketing Management, 43*(4), 721–725.
Zahra, S. A., & George, G. (2002). Absorptive capacity: A review, reconceptualization, and extension. *Academy of Management Review, 27*(2), 185–203.
Zajac, E. J., Kraatz, M. S., & Bresser, R. K. F. (2000). Modelling the dynamics of strategic fit: A normative approach to strategic change. *Strategic Management Journal, 21*(4), 429–453.
Zhang, Y., & Li, H. (2010). Innovation search of new ventures in a technology cluster: The role of ties with service intermediaries. *Strategic Management Journal, 31*(1), 88–109.
Zhang, C., Wu, F., & Henke, J. W. (2015). Leveraging boundary spanning capabilities to encourage supplier investment: A comparative study. *Industrial Marketing Management, 49*, 84–94.
Zheng, W., Singh, K., & Mitchell, W. (2015). Buffering and enabling: The impact of interlocking political ties on firm survival and sales growth. *Strategic Management Journal, 36*(11), 1615–1636.

Lin Cui (Ph.D *The Australian National University*) is an associate professor in international business in the Research School of Management of the Australian National University (ANU). His research centres on emerging market firms' strategies and operations, with special focus on Chinese firms. He has broad research interests in internationalization, corporate governance, knowledge transfer, and cross-cultural management. His recent studies are published in *Journal of International Business Studies, Organization Studies*, *Journal of World Business, Management International Review, International Business Review, Asia Pacific Journal of Management,* and *Journal of Business Research*.

Di Fan* (Ph.D *Monash University*, CPA) is an associate professor in international business in the School of Management of Curtin Business School at Curtin University, Perth, Australia. His current research interest includes, Business Strategies, and Multinational Firms. His publications appear in journals, such as *Organization Studies*, *Technovation*, *Academy of Management Learning & Education*, *Journal of World Business*, *International Business Review*, *Global Strategy Journal*, and *Journal of Business Ethics* etc.

Feng Guo (PhD Candidate, *Harbin Institute of Technology, The Australian National University*) is a Ph.D. candidate at School of Management, Harbin Institute of Technology (HIT) in China, and a visiting scholar in international business in the Research School of Management of the Australian National University (ANU). His current research interests are innovation management and strategy management.

Yi Fan (Research Candidate,*The Australian National University*) is a research candidate in international business in the Research School of Management of the Australian National University (ANU). His research centres on Internationalization Choice of Chinese Firms.